

Sequences & Series



The binomial series for negative and fractional powers

- 1 Expand $(1 + x)^{-1}$ up to the term in x^3 , using $(1 + x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3$.
 - 2 Expand $(1 + x)^{1/2}$ up to the term in x^3 .
 - 3 Expand $(1 - x)^{-3}$ up to the term in x^2 .
 - 4 Expand $(1 + x)^{-4}$ up to the term in x^2 .
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Validity conditions for the binomial series

- 5 State the range of values of x for which the binomial expansion of $(1 + x)^{-2}$ is valid.
 - 6 State the range of values of x for which the expansion of $(3 + x)^{-1}$ converges, given that after factoring, the general form requires $\left|\frac{x}{3}\right| < 1$.
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Using series expansions to approximate values

- 7 Use $(1 + x)^{1/2} \approx 1 + \frac{x}{2} - \frac{x^2}{8}$ to approximate $\sqrt{1.02}$, to four decimal places.
 - 8 Use $(1 + x)^{-1} \approx 1 - x + x^2 - x^3$ to approximate $\frac{1}{1.05}$, to four decimal places.
 - 9 Use the geometric series $(1 - x)^{-1} \approx 1 + x + x^2 + x^3$ (the binomial series with $n = -1$) to approximate $\frac{1}{0.98}$, to four decimal places.
 - 10 Use $(1 + x)^{-1} \approx 1 - x + x^2$ to approximate $\frac{1}{1.01}$, to four decimal places.
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Expansions of the form $(a+bx)^n$

- 11 Expand $(4 + x)^{1/2}$ up to the term in x^2 , by writing $(4 + x)^{1/2} = 4^{1/2} \left(1 + \frac{x}{4}\right)^{1/2}$.
- 12 Expand $(2 - 3x)^{-1}$ up to the term in x^2 , by writing $(2 - 3x)^{-1} = 2^{-1} \left(1 - \frac{3x}{2}\right)^{-1}$, and state the range of validity.
- 13 Expand $(1 + 2x)^{-2}$ up to the term in x^2 , and state the range of validity.

- 14 Expand $(9 - x)^{1/2}$ up to the term in x^2 , by writing $(9 - x)^{1/2} = 9^{1/2} \left(1 - \frac{x}{9}\right)^{1/2}$.
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Constructing Maclaurin series

- 15 Find the Maclaurin series for $f(x) = \cos x$ up to the term in x^4 , using $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4$.
- 16 Find the Maclaurin series for $f(x) = e^{2x}$ up to the term in x^3 .
- 17 Find the Maclaurin series for $f(x) = \ln(1 - x)$ up to the term in x^3 .
- 18 Find the Maclaurin series for $f(x) = \sin(2x)$ up to the term in x^3 , using $\sin u \approx u - \frac{u^3}{6}$ with $u = 2x$.
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Building new series from known ones

- 19 Using $e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$, find the Maclaurin series for e^{-x^2} up to the term in x^4 , by substituting $-x^2$ for x .
- 20 Given $\sin x \approx x - \frac{x^3}{6}$ and $(1 + u)^{-1} \approx 1 - u + u^2$, find the Maclaurin series for $\frac{1}{1 + \sin x}$ up to the term in x^2 (using $u = \sin x \approx x$ to this order).
- 21 Using $\frac{1}{1 - x} \approx 1 + x + x^2 + x^3$, differentiate term by term to find the Maclaurin series for $\frac{1}{(1 - x)^2}$ up to the term in x^2 .
- 22 Using $\ln(1 + x) \approx x - \frac{x^2}{2} + \frac{x^3}{3}$, find the Maclaurin series for $\ln(1 + 2x)$ up to the term in x^3 , by substituting $2x$ for x .
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Using series to approximate definite integrals

- 23 Use $\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$ to approximate $\int_0^{0.5} \cos x \, dx$, to four decimal places, and compare with the exact value $\sin(0.5) \approx 0.4794$.
- 24 Given the standard result $(1 - x^2)^{-1/2} \approx 1 + \frac{x^2}{2} + \frac{3x^4}{8}$, approximate $\int_0^{0.2} (1 - x^2)^{-1/2} \, dx$, to four decimal places, and compare with the exact value $\arcsin(0.2) \approx 0.2014$.
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A well-designed word problem: relativistic momentum correction

- 25 This question has three parts. In special relativity, the momentum correction factor is $\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2}$, where v is speed and c is the speed of light. (a) Using the binomial expansion with $u = -\frac{v^2}{c^2}$, show that $\gamma \approx 1 + \frac{v^2}{2c^2} + \frac{3v^4}{8c^4}$ for small $\frac{v}{c}$. (b) A satellite travels at $v = 7000$ m/s, with $c = 3 \times 10^8$ m/s. Using only the first-order approximation $\gamma \approx 1 + \frac{v^2}{2c^2}$, find $\gamma - 1$, to three significant figures, in standard form. (c) Explain, by comparing orders of magnitude, why the second term $\frac{3v^4}{8c^4}$ is negligible for this satellite compared to the first-order term.